

Trust in Neural Intrusion Detection: Auditing Post-hoc Explanation Reliability

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Abstract—Post-hoc explanations can support audits of neural intrusion detectors, but generating an attribution does not establish that it is reliable evidence. We examine reproducibility, functional response and outcome dependence on corrected CIDS2017 and NF-ToN-IoT-V2 binary classification, using MLP and 1D-CNN models, five training seeds and five explainers. We separate stochastic recomputation, numerical sensitivity and deterministic controls, reporting identifiable coverage alongside rank and top-10 agreement. Functional response compares attribution-guided deletion/insertion with displacement-matched random controls on a common panel under three replacement operators. LIME rank correlations range from .766 to .879; high conditional agreement for other methods can conceal non-identifiable top-10 selections. For ToN-IoT, IG insertion effects change from positive under median replacement to negative under conditional Gaussian replacement in both architectures. An exploratory correct/error analysis is limited to three deterministic methods and explicitly reports scarce false-negative support. The results support condition-specific auditing of frozen models, not an operator-independent ranking of explainers, validated alert triage, or a guarantee of IDS safety.

Index Terms—explainable artificial intelligence, intrusion detection, feature attribution, explanation evaluation, neural networks

I. INTRODUCTION

A neural intrusion detection system (IDS) can perform well on held-out traffic without establishing that its feature attributions are reliable evidence for auditing individual decisions. An analyst inspecting a flagged flow needs to know what an explanation measures, whether it can be reproduced, and whether the attributed features affect the model under an explicit intervention. Trust motivates these questions; neither predictive accuracy nor the presence of an explanation answers them. Benchmark performance also need not transfer to the conditions in which an IDS is deployed [1].

Post-hoc methods produce different accounts of a fitted model. A local surrogate coefficient, a path-integrated gradient and a replacement effect are not interchangeable measurements. Repeating an account can establish reproducibility without establishing that its ordering predicts a functional response. Conversely, a deletion experiment depends on what replaces the deleted coordinates [2], [3]. An audit must there-

fore separate properties of the explanation from choices made by its evaluator.

We ask: *To what extent do post-hoc explanations provide reliable evidence for auditing neural intrusion detection decisions?* We operationalise reliability through three questions, not a composite trust score: **RQ1**: how reproducible are attributions under repeated explanation sampling or changes in numerical integration, for a fixed model and target? **RQ2**: how do attribution-guided interventions compare with matched random controls, and how does this response depend on the replacement operator? **RQ3**: how does the response vary across correct and incorrect decisions, including false positives and false negatives?

The study reorganises existing experiments on two binary traffic benchmarks, two neural architectures, five training seeds and five explainers. Its contribution is a condition-resolved audit: metric-specific coverage accompanies reproducibility; paired functional effects are evaluated under three distinct operators; and an exploratory outcome analysis states its restricted method coverage and sample support. We evaluate evidence about frozen models, not analyst trust, incident reduction, or a criterion for declaring an IDS safe.

II. RELATED WORK

Warnecke et al. evaluate explanation accuracy, sparsity and security-oriented criteria, including efficiency, for malware and vulnerability models [4]. Kumar and Thing compare explanations of deep network intrusion detectors using agreement and security-focused criteria [5]. Maseno et al. assess SHAP through permutation importance, retraining, feature-removal and spurious-feature diagnostics on ToN-IoT and UNSW-NB15, using logistic regression, random forests and XGBoost [6]. Our contribution is incremental to these multidimensional evaluations; different models, targets and interventions preclude importing their numerical method rankings.

Evaluation procedures themselves require scrutiny. MetaQuantus examines the behaviour of explanation-quality estimators [7]; ROAD and deletion-benchmark analyses expose dependence on removal rules and induced inputs [2], [3]. We address this dependence directly by retaining fixed accounts and targets while changing replacement operators. We also distinguish independently resampled LIME/KernelSHAP accounts from IG quadrature sensitivity and deterministic repetition. This prevents several different

This work was partially funded by “CyberPRAISE – Cybersecurity research for PRivAte, Intelligent and truStAbLe solutions” (NORTE2030-FEDER-01820300), supported by the European Regional Development Fund through NORTE2030 under the PORTUGAL 2030 Partnership Agreement.

TABLE I

BINARY PROTOCOL. WIDTHS DENOTE HIDDEN UNITS OR CONVOLUTION FILTERS; THE CNN DENSE WIDTH FOLLOWS THE SLASH. SEEDS 42–46 SHARE ONE DATA SPLIT. BUDGETS ARE EXPERIMENTAL SETTINGS, NOT MEASURED SERVICE LATENCY.

	CICIDS2017 (80 features)	NF-ToN-IoT-V2 (40 features)
Train / validation / test	349,972 / 71,735 / 71,364	350,000 / 74,998 / 74,998
MLP widths; dropout; batch	256–128–64; .20; 512	256–128; .10; 256
CNN filters/dense; dropout; batch	64–128–256/128; .20; 512	32/64; .10; 512
Explainer	Configuration	Reproducibility test
Gradient×Input (G×I)	Gradient times standardised input	No stochastic estimate
Integrated Gradients (IG)	Training median; 64 trapezoid intervals	64 versus 128 intervals
Occlusion	One coordinate replaced by training median	Fixed-rule repetition
LIME	32,000 perturbations; Ridge; all features	50 independently seeded runs
KernelSHAP (SHAP)	102,400 coalitions; identity link; median	Two independently seeded runs

notions of consistency from becoming a single ranking of explainers.

III. EXPERIMENTAL PROTOCOL

A. Data, Models and Operating Points

The tasks classify benign versus attack flows from the corrected `CICIDS2017_improved` re-extraction [8], [9] and the `NF-ToN-IoT-V2` NetFlow release [10]. Table I specifies the retained matrices and selected models. Preparation removes non-finite rows, clips features at 10^{12} and caps each class at 250,000 rows. Exact clipped `float64` feature hashes define groups; conflicting-label groups are removed before whole-group allocation to one fixed 70/15/15 split. Training-only transformations perform imputation, variance filtering and standardisation. A label-blind guard removes final `float32` validation/test representations matching an earlier partition exactly or after five-decimal rounding: 3,260/3,630 CICIDS rows and 2/2 ToN-IoT rows. This checks representation overlap, not independence of sessions or time.

Validation selects at most four configurations per task/model at seed 1701; selected models are fitted at seeds 42–46. MLPs use ReLU/dropout hidden layers; CNNs use width-three convolutions, batch normalisation, max-pooling by two and a ReLU/dropout dense layer. CNN neighbourhoods follow the recorded feature order, not a natural traffic sequence. All models use a sigmoid output, binary cross-entropy, Adam (10^{-3} initial learning rate), capped class weights and validation-loss stopping. Final fitting permits 50 epochs with patience seven and restored best weights. The decision threshold minimises validation false negatives subject to $\text{FPR} \leq .05$; test data do not select it.

B. Explainers, Targets and Panels

We use Gradient×Input, IG, Occlusion, LIME and KernelSHAP [11]–[14]. The first two require gradients; the others query model outputs. Zero in the standardised input is the training-mean origin, not absent traffic. LIME fits all continuous features without discretisation, using Gaussian perturbations about a training-background mean, Euclidean distance, kernel width $.75\sqrt{d}$ and Ridge penalty one. Each run resamples its class-stratified background (cap 50,000 rows),

so its recomputation includes background variability. LIME coefficients are not reference-relative contributions. SHAP uses one median point repeated as 100 physical background rows, not 100 distinct cases. Historical calls omit `ll_reg`; per-stage library versions were not recorded. The observed sparse accounts are consistent with `num_features(10)`, the default since SHAP 0.47, but this does not recover each run’s effective setting. SHAP results therefore describe the stored accounts, not a reproducible claim for a fully specified configuration.

Let $s_c(x)$ be the fitted probability of class c . The target remains fixed through each explanation/intervention. The *central* panel selects up to 128 flows per true class/seed, correctly classified by both models; the *stochastic* subset selects eight. Each task supplies 1,280 central source–seed records and 80 stochastic records, evaluated by both models. CICIDS/ToN-IoT central panels contain 1,276/1,269 distinct source IDs; each stochastic panel contains 80. Selection does not read attributions. RQ1 uses central records for IG/Occlusion and the subset for LIME/SHAP. RQ2 uses the same 80 records and targets for every method, model and operator within a task. LIME’s RQ2 account averages its 50 coefficient vectors; SHAP uses its primary run. These are not per-run functional uncertainty estimates.

RQ3 uses a separate model-specific *outcome* panel, capped at 128 flows per true class, correctness status and seed. It contains 3,897 CICIDS and 5,120 ToN-IoT model–seed records. Only G×I, IG and Occlusion were evaluated. The target is the predicted class: attack for TP/FP and benign for TN/FN. Thus, comparisons between TP and FN change target as well as outcome.

C. Reproducibility and Identifiability (RQ1)

Accounts are ranked by decreasing absolute attribution. We report Spearman correlation ρ and top-10 Jaccard J_{10} , conditional on identifiable pairs, alongside their separate coverage fractions. With $\epsilon = \max(10^{-12}, 10^{-8} \max_j |\phi_j|)$, an absolute-attribution range $\leq \epsilon$ invalidates the rank comparison; a top-10 boundary gap $\leq \epsilon$ invalidates only the set comparison. Non-identifiable pairs are not observed zero correlations. Top-10 covers 12.5% of CICIDS coordinates but 25% of ToN-IoT

TABLE II

RQ1, CONDITIONAL SIMILARITIES WITH 95% INTERVALS AND METRIC-SPECIFIC COVERAGE C (%). N : AVAILABLE SOURCE-SEED RECORDS; S : DISTINCT SOURCES. R = STOCHASTIC RECOMPUTATION; Q = QUADRATURE SENSITIVITY; D = DETERMINISTIC CONTROL. LIME AVERAGES 1,225 DEPENDENT RUN PAIRS PER RECORD, NOT 1,225 INDEPENDENT OBSERVATIONS. SHAP INTERVALS CONDITION ON TWO RUNS.

Condition	Method/test	N	S	ρ [95% CI]	C_ρ	J_{10} [95% CI]	C_J
CICIDS/MLP	LIME/R	80	80	0.879 [0.870, 0.889]	100.0	0.634 [0.614, 0.653]	100.0
	SHAP/R	80	80	0.980 [0.956, 0.995]	100.0	0.959 [0.924, 0.988]	76.2
	IG/Q	1280	1276	0.9997 [0.9996, 0.9997]	100.0	0.989 [0.984, 0.993]	100.0
	Occl./D	1280	1276	1.000 [1.000, 1.000]	69.0	1.000 [1.000, 1.000]	47.0
CICIDS/CNN	LIME/R	80	80	0.792 [0.775, 0.808]	100.0	0.608 [0.543, 0.660]	100.0
	SHAP/R	80	80	0.975 [0.956, 0.990]	82.5	0.943 [0.905, 0.978]	71.2
	IG/Q	1280	1276	0.9988 [0.9985, 0.9991]	100.0	0.964 [0.958, 0.970]	100.0
	Occl./D	1280	1276	1.000 [1.000, 1.000]	83.6	1.000 [1.000, 1.000]	46.1
ToN-IoT/MLP	LIME/R	80	80	0.767 [0.749, 0.783]	100.0	0.670 [0.643, 0.699]	100.0
	SHAP/R	80	80	0.998 [0.992, 1.000]	100.0	0.985 [0.945, 1.000]	45.0
	IG/Q	1280	1269	0.9996 [0.9995, 0.9997]	100.0	0.986 [0.982, 0.990]	87.2
	Occl./D	1280	1269	1.000 [1.000, 1.000]	100.0	1.000 [1.000, 1.000]	58.6
ToN-IoT/CNN	LIME/R	80	80	0.766 [0.739, 0.791]	100.0	0.638 [0.601, 0.680]	100.0
	SHAP/R	80	80	0.999 [0.994, 1.000]	100.0	0.996 [0.980, 1.000]	50.0
	IG/Q	1280	1269	0.9995 [0.9992, 0.9997]	100.0	0.983 [0.976, 0.989]	87.2
	Occl./D	1280	1269	1.000 [1.000, 1.000]	100.0	1.000 [1.000, 1.000]	77.1

coordinates. LIME/SHAP test resampling, IG tests numerical approximation, and Occlusion tests deterministic implementation consistency. Neither identical repetition nor a 128-step approximation is ground truth. For IG we also inspect absolute completeness residuals, $|\sum_j \phi_j - [s_c(x) - s_c(r)]|$, separately from rank agreement.

D. Functional Response and Operators (RQ2–RQ3)

For a fixed magnitude ranking, select the first $\lceil ud \rceil$ coordinates, where $u \in \{.05, .10, .20, .30, .50\}$. Deletion replaces selected coordinates; insertion retains them and replaces their complement. Effects are $e_{\text{del}}(u) = s_c(x) - s_c(x_{\text{del}}(u))$ and $e_{\text{ins}}(u) = s_c(x_{\text{ins}}(u)) - s_c(r_o)$, where r_o is the operator’s complete-replacement input. For each arm we compute

$$\Delta\text{AOPC} = \frac{1}{0.5} \int_0^{0.5} [e_{\text{attribution}}(u) - \bar{e}_{\text{random}}(u)] du, \quad (1)$$

using trapezoids and a zero origin. For each flow/fraction, 50 frozen random subsets match attributed-set counts in five displacement-rank bins, sampling coordinates without replacement. Matching controls subset size and displacement bins, not exact displacement or traffic feasibility. Controls are method-specific; comparing adjusted effects is not comparing raw attribution effects.

The historical stable sort breaks exact magnitude ties by increasing feature index. RQ2 retains all 80 records, including constant accounts and boundaries non-identifiable at the RQ1 tolerance. We audit each fraction and repeat the functional calculation with reversed and three seeded random orders within tied blocks, re-matching controls with the same seed streams. On the same RQ2 panel, we compare IG at 64 and 128 steps and double further until every record of a fitted model has completeness residual $\leq .01$ or 2,048 steps are reached; no record is excluded at the cap.

Three RQ2 operators represent different replacement mechanisms: *training median* replaces coordinates independently by marginal medians; *conditional Gaussian mean* uses a ridge-regularised training covariance to estimate replaced coordinates conditional on retained ones; *joint donor* copies a replaced block from one sampled training row. The latter preserves within-block dependence, not compatibility with retained coordinates. Covariance uses the first 100,000 training rows plus $10^{-6}I$; the donor pool samples up to 50,000 training rows without replacement. Displacement bins rank $|x - r_o|$, using marginal medians, the Gaussian mean, or the sampled complete donor, respectively. Conditional insertion recomputes the complement given the inserted features; it is not a path from a fixed median. The accounts remain fixed across operators. This exploratory selection represents procedural diversity among nine cached operators, not a search for the largest reversals. RQ3 uses only training-median replacement. Magnitude rankings combine positive and negative contributions: a negative ΔAOPC is not automatically a wrong explanation.

E. Aggregation and Scope of Uncertainty

We average within each source/seed, then weight the five fitted seeds equally. The 95% intervals cross-resample source identities and seeds, preserving source reuse; new conditional estimates and within-source operator contrasts use 5,000 replicates. LIME intervals additionally retain the shared-run jack-knife treatment of its dependent pairwise comparisons. SHAP intervals are conditional on two observed runs, not an estimated run population. Existing cell intervals are retained only after independently reconstructing the identical estimand and support. No interval is obtained by averaging previous bounds. Functional intervals condition on frozen accounts and random controls; they exclude fresh explainer/operator-draw variability. Comparisons are exploratory, with marginal rather than

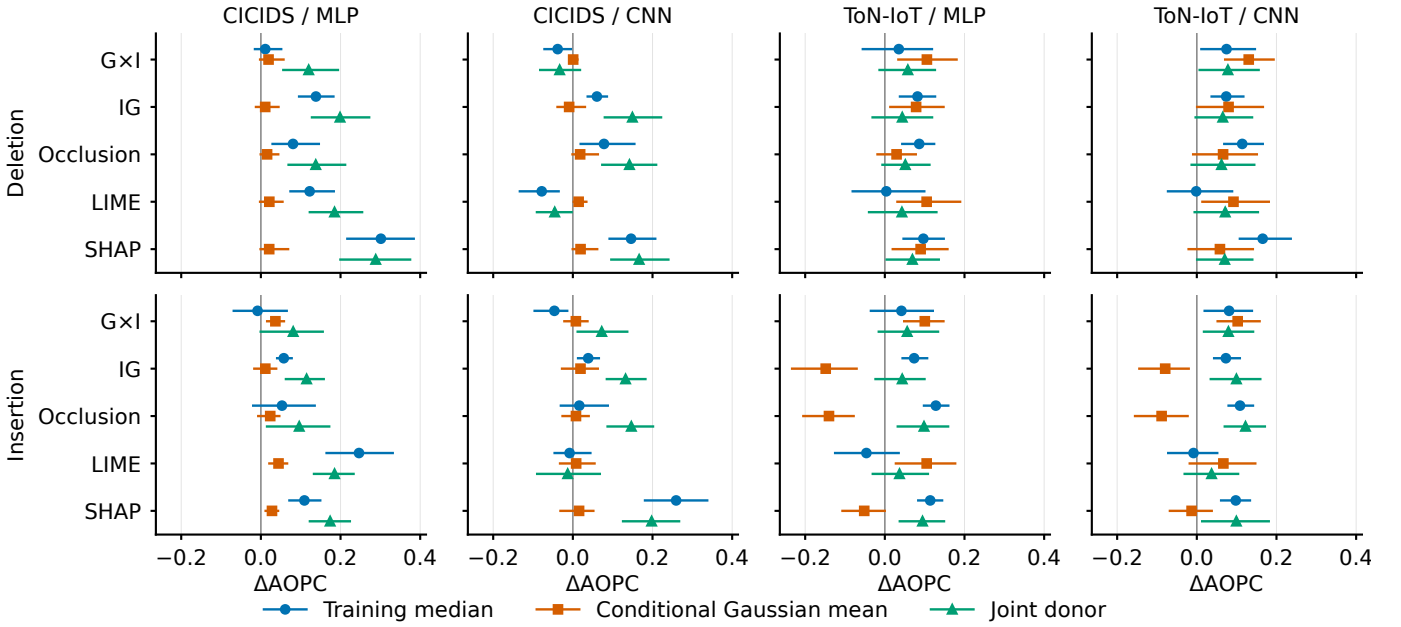


Fig. 1. RQ2: attribution-minus-displacement-matched-random AOPC for deletion (top) and insertion (bottom). Points are equal-seed means; bars are marginal 95% source/seed intervals. Each task uses the same 80 distinct sources across all methods, models and operators (16 per seed). Accounts and targets stay fixed. Intervals condition on stored accounts, feature-index tie breaking and intervention draws; the three operators were selected for their different replacement mechanisms.

multiplicity-adjusted intervals. The version-matched repository package records source/seed identities, configurations, hashes, exclusions and paired contrasts. Its reproduction instructions rebuild Table II and both figures from the included per-source records without model calls. A separate command reconstructs the sensitivity summaries from their source-level measurements. The package distinguishes recorded, reconstructed and unknown SHAP provenance. Multiclass results do not enter the reported aggregates.

IV. RESULTS

A. Predictive Context

Across five seeds, mean test F1 is .9798 (MLP) and .9821 (CNN) on CICIDS, and .9740/.9728 on ToN-IoT. Corresponding mean FPRs are .0445/.0395 and .0492/.0490. These operating points motivate auditing well-performing models; class caps and one benchmark split do not establish deployment precision.

B. RQ1: Reproducibility Is Property-Specific

Table II separates conditional agreement from coverage. LIME’s mean rank correlation ranges from .766 to .879, with top-10 agreement .608–.670 and complete coverage. Its finite-budget recomputation is therefore not interchangeable with a deterministic account. The stored SHAP pairs’ conditional rank correlations are .975–.999, but CICIDS/CNN rank coverage is only 82.5%. Top-10 coverage is lower still: 45.0–76.2% across conditions. A high conditional agreement does not ensure that the requested ranking or top- k selection is

identifiable throughout the panel. Two runs cannot establish arbitrary-run robustness.

IG’s 64-to-128-step rank correlation exceeds .998 in every condition. Conditional top-10 agreement is .964–.989; ToN-IoT top-10 coverage is 87.2%. Rank convergence is not completeness: at 64 steps the per-condition 95th percentile absolute completeness residual ranges from .036 to .193 probability units. Occlusion’s identifiable repeated accounts agree exactly, as expected; its top-10 coverage ranges from 46.1% to 77.1%. This deterministic control validates repeatability of a rule, not its explanatory correctness. Gradient×Input has no stochastic-variability estimate in this protocol.

C. RQ2: Functional Evidence Depends on the Operator

Figure 1 shows effects, rather than an aggregate method rank. With median replacement, the stored SHAP accounts have positive mean deletion responses in all four conditions, whereas LIME changes from .123 on CICIDS/MLP to $-.078$ on CICIDS/CNN and is near zero on ToN-IoT. With conditional replacement, CICIDS deletion effects concentrate nearer zero across methods. Joint-donor replacement produces yet another pattern; for example, CICIDS/CNN IG deletion increases from .060 (median) to .149 (donor), with a paired change of .089 [95% CI .025, .159].

The insertion arm makes operator dependence particularly explicit. ToN-IoT IG changes from .074 [.045, .106] under median replacement to $-.148$ [$-.233$, $-.071$] under the conditional mean for the MLP, and from .073 [.044, .108] to $-.079$ [$-.144$, $-.020$] for the CNN. The paired conditional-minus-median changes are $-.222$ [$-.318$, $-.131$] and $-.152$ [$-.210$,

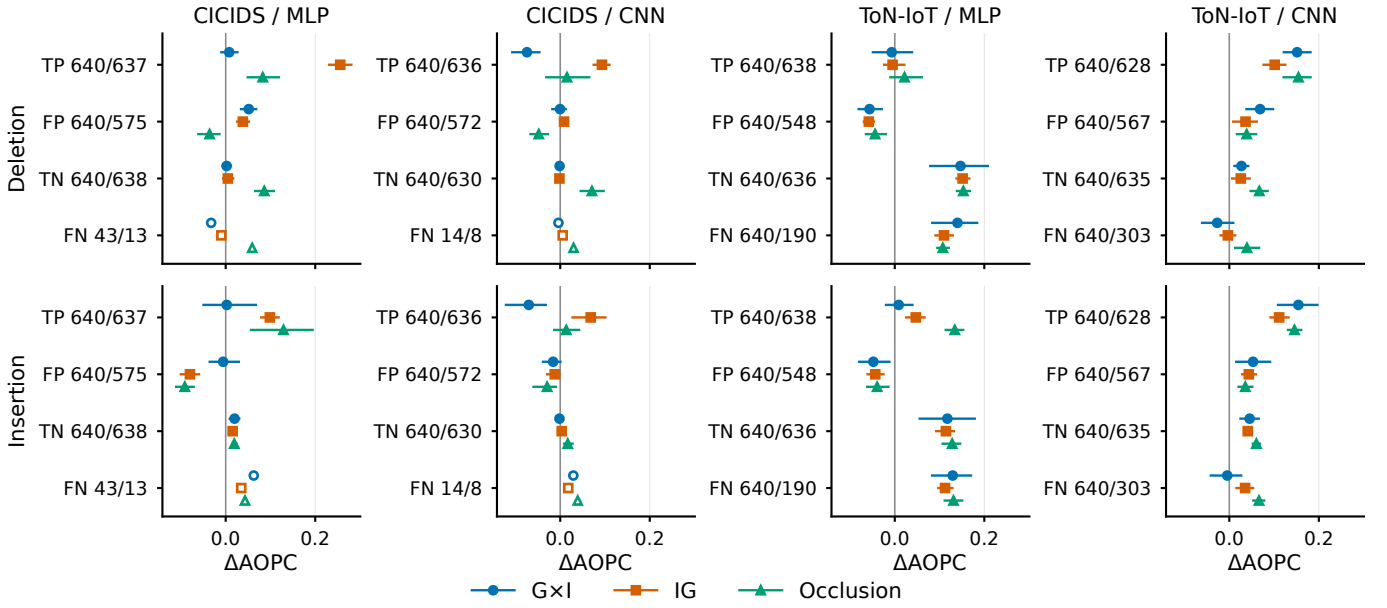


Fig. 2. RQ3: training-median functional response on model-specific outcome panels, separately for deletion and insertion. Labels give source-seed records/distinct sources; methods within a condition share them. Bars show marginal 95% source/seed intervals. Open points without bars denote low-support CICIDS false negatives, retained only descriptively. Targets are attack for TP/FP and benign for TN/FN. No averaging across methods, architectures or outcome strata is used.

−.095], respectively. These compare identical source-seed records and fixed targets; they are not differences inferred from overlap of separate intervals. By contrast, the corresponding IG deletion changes are −.003 [−.080, .075] and .006 [−.084, .098], compatible with zero. The operator dependence is thus arm-specific as well as condition-specific.

Every stored SHAP selection with $k > 10$ has a non-identifiable boundary (100% in each condition); the artifact tabulates all fractions and methods. Reversing tied blocks changes training-median deletion for SHAP by −.041 on CICIDS/MLP and insertion for Occlusion by .095 on CICIDS/CNN. Thus, functional effects depend on tie breaking as well as the operator. The ToN-IoT IG insertion contrast remains negative under all four alternative orders.

At 128 steps, the ToN-IoT IG insertion contrasts are −.214 (MLP) and −.147 (CNN). At the final resolution they are −.211 [−.311, −.123] and −.147 [−.200, −.094], respectively (95% CIs). All 320 RQ2 model-source records reach absolute completeness residual $\leq .01$, with 256–2,048 steps across fitted conditions. Per-source contrasts and initially high-residual cases remain in the artifact. The highlighted operator contrast therefore persists under these numerical and tie-order checks; completeness alone does not certify the ranking.

These findings concern a method-operator-target combination. Conditional replacement asks a different intervention question, and magnitude ordering does not encode whether each feature supports or opposes the target score. Neither a negative adjusted effect nor a rank reversal alone identifies a defective explanation. The evidence does rule out treating this evaluation as an operator-independent endorsement of a

method.

D. RQ3: Outcome-Stratified Diagnostics

Figure 2 restricts the analysis to the three evaluated methods and reports support beside each stratum. Outcomes do not yield a consistent ordering across models: with IG, ToN-IoT false-positive deletion response is −.058 on the MLP but .036 on the CNN. Large responses can accompany errors, and small responses can accompany correct decisions; these group summaries do not estimate whether an individual alert is wrong.

Support also matters. CICIDS false negatives comprise 43 model-seed records from 13 distinct sources for the MLP, and 14 from eight for the CNN. They remain descriptive; no inferential whiskers are shown for those strata. The outcome panels are sampled, not estimates of error prevalence.

Score confounding limits comparisons even with a fixed attack target. Mean attack thresholds are .00046/.00025 for CICIDS MLP/CNN and .173/.265 for ToN-IoT: an attack decision need not imply attack probability above .5. An exploratory within-seed FP-TP attack-score match (common score range, one-to-one assignment, .2-standard-deviation caliper) retains only ten CICIDS/MLP pairs and three CICIDS/CNN pairs, insufficient for the prescribed interval procedure. ToN-IoT retains 129/138 pairs, describing only common score support. The unadjusted figure is not an error effect adjusted for confidence. Matching TP to FN cannot resolve this limitation because it changes the explained target. No LIME/SHAP outcome evidence or validated alert-triage mechanism is inferred.

TABLE III

SINGLE-INPUT CPU TIMINGS (MS); FIRST REPETITION EXCLUDED. TWO FIXED INPUTS SUPPLY THE n INDIVIDUAL MEASUREMENTS PER METHOD.

Method	Median	Min-max	n
G×I	4.4	3.7–4.6	8
IG	238.5	233.5–256.9	8
Occlusion	166.3	163.3–191.4	8
LIME	171.8	160.2–174.6	6
SHAP	988.3	899.5–1,101.1	6

V. DISCUSSION AND LIMITATIONS

The audit supports a qualified use of post-hoc explanations: they can provide reproducible accounts and measurable evidence about frozen models, provided the account’s target, identifiable support and evaluation rule are specified. Reproducibility does not imply that an intervention will outperform a matched random subset. Conversely, an intervention response describes the model under that replacement mechanism, not a feature’s causal role in an attack. The practical reporting unit should therefore retain the method, target, reference, operator, panel and uncertainty together. Collapsing these into a trust score would hide distinctions that the experiments expose.

Computational scope. A cached two-input CICIDS/MLP microbenchmark on an Apple M1 Pro CPU (10 cores, 32 GiB), excluding the first repetition and preparation, measures one explanation of one input per observation, not a joint two-input batch. Table III reports the median, range and measurement count. These are small-sample descriptive timings: gradient/Occlusion batches were one, and LIME/SHAP prediction batches 4,096. SHAP used one physical background row, unlike the main audit’s 100 copies. One explanation is not LIME’s 50-run recomputation experiment or the full intervention audit. Historical batch-stage costs include preparation and are not pooled with these timings. Full-audit critical-path time and operational per-alert latency remain unknown.

Limits. One fixed split and five fitted seeds do not quantify new-site, temporal or session-level generalisation. Exact/rounded overlap safeguards do not establish complete independence; source identity is also not a session cluster. Class caps alter prevalence, and the correct-by-both panel conditions RQ1/RQ2 on successful decisions. The processed ToN-IoT extract lacks timestamps and endpoint addresses needed for stronger split policies. Feature-space replacements need not describe feasible traffic. The CNN’s feature order and selected configurations limit architectural interpretation.

The RQ structure and operator/outcome extensions reorganise existing experiments; sampling budgets were increased after inspecting earlier results. They are not retroactively preregistered hypotheses. SHAP’s two-run support and unknown historical regularisation/environment further limit reproducibility claims. The explicitly configured SHAP 0.51.0 microbenchmark does not recover the historical runs’ settings. Marginal intervals and exploratory example selection do not establish a universal method winner. The study measures nei-

ther analyst understanding/trust nor detection gains from using explanations, and proposes no operational safety threshold.

VI. CONCLUSION

For the evaluated binary tasks and fitted pipelines, post-hoc attributions provide condition-specific evidence about model behaviour, but do not independently validate the correctness or safety of IDS decisions. Identifiable accounts can be highly reproducible while functional conclusions change with the replacement operator. Outcome diagnostics expose further dependence on the model, score and available support. An auditable explanation must therefore state these conditions; explaining a decision is not equivalent to validating an IDS.

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